

When predicting the future values for the house price index, I developed four models using regression. When constructing the forecast for the index, I chose a linear time trend to be applied in the equation. In comparison to a quadratic equation, they both struggled to follow the larger trends in the actual data. The equation shown in figure one and two considers the passage of time. Each quarter that passes is counted as additional “t” and it is multiplied by the slope and added with y.

Along with the time trend, I added State Tax Collection Totals for California as my additional independent variable. I decided on this one as it

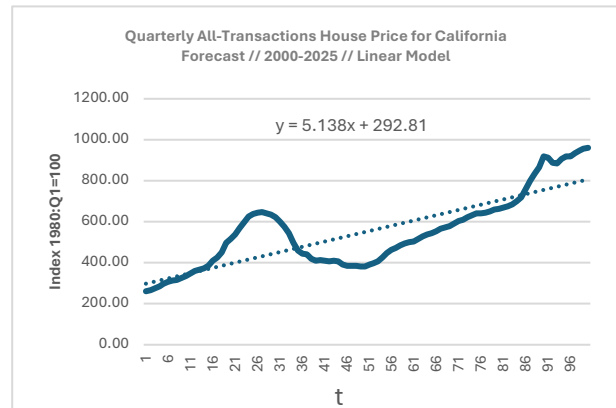


Figure 1

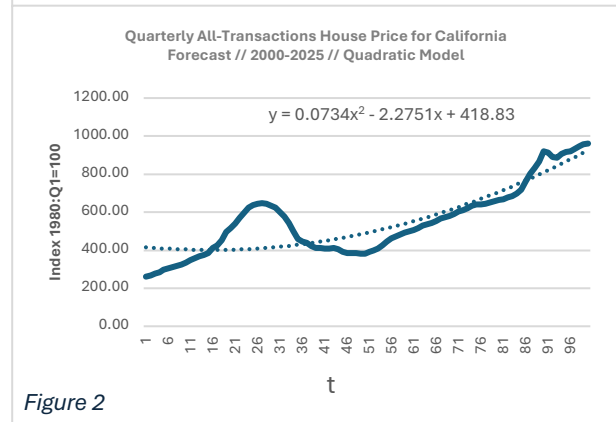


Figure 2

could be a decent factor into the prices of housing in the state. Using the “CORREL” function in Excel, the readings of 0.814; show a strong positive correlation between the sets of data. Another aspect as to why I chose it is because the data also covers California and I think the forecast can be improved with a localized variable. A concern when using this data are the seasonal peaks it has around the second quarters, which may result in over forecasting some time periods.

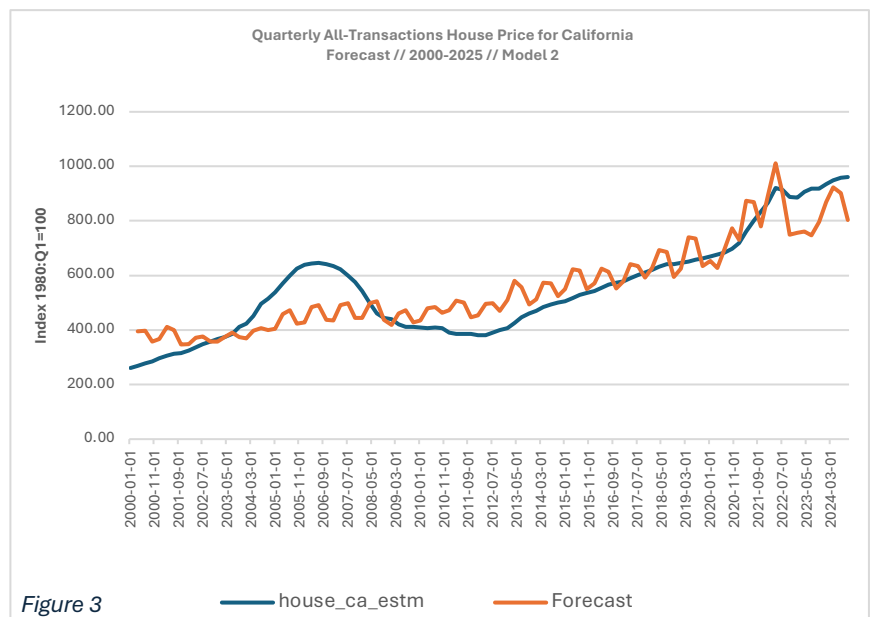
The first model introduced has time trend as the only independent variable in the regression. It is a linear equation with an upward trend that matches the overall movement of the quarterly house price index, our dependent variable. (Figure 1) It does not account for the cycles that occur in the actual data and either under or over forecasts as a result. The equation that is used to predict the dependent variable is $292.807 + 5.138(t)$. That is, for every one unit increment of time, it is multiplied by 5.138 and is added with 292.807. The regression analysis shows the p-value being less than alpha so I can conclude that the time variable does have a relationship with the house index. When analyzing the closeness of the predicted

data versus the actual results, the indicators present high values of 108.46 for the RMSE and the 91.23 for the MAD. The readings from the r-squared is 0.65, which means that 65% of the variations of the house price index can be explained with time series as an independent variable.

The second model adds two new variables which are the State Tax Collections for California and a lagged set of that same variable by just one period in the past. The lagged variable will aid in the forecast by capturing any delayed effect of the index. The equation for predicting the house price index is $177.5214 + 0.0882(t) + 0.0043(\text{tax_ca}) + 0.0047(\text{lag1_tax_ca})$. The translation for the tax variables is that for each increase in a million dollars in tax collections, the house price index increases by 0.0882 and 0.0043%. When measuring the closeness of forecast, it improves its accuracy slightly when compared to the first model. The RMSE drops down to 89.72 and the MAD decreases to 73.35. The r-squared increases to a measure of 75.79% for the variations in the index that can be explained with the added variables. This will be shaved down to 75% with the adjusted r-squared to account for the additional variables in the forecast.

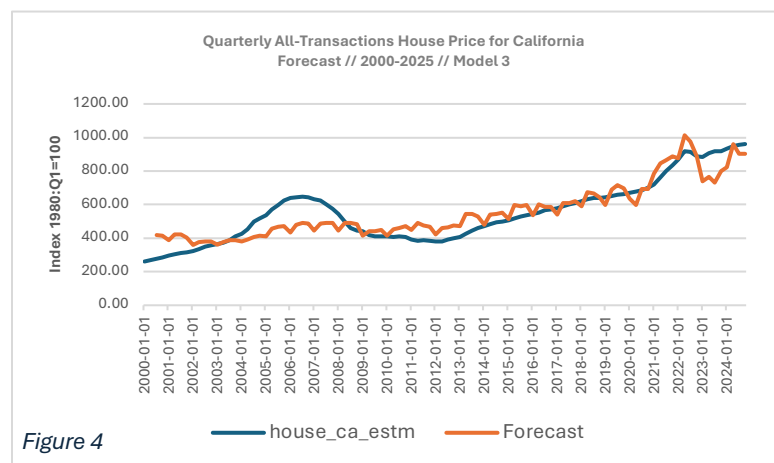
When plotted on a time graph, the seasonal spikes from the tax collection data are noticeable. Which was my concern with using this data as it would over forecast patterns that are not reflected in the index.

(Figure 3) The regression analysis shows the p-value being above the alpha of 0.05 for the time variable. This would make the factor have no relationship with the index for this model. The confidence intervals of the time trend also include zero and it adds to the case that it does not support the relationship also.



The third model has all the variables from before but also adds another lagged set for two time periods in the past in tax collections. The equation for the predicted index is $145.7538 - 0.6021(t) + 0.0039(\text{tax_ca}) + 0.0040(\text{lag1_tax_ca}) + 0.0041(\text{lag2_tax_ca})$. The one unit increases for these variables are the same as before just with the added lag period. The discussion is also similar in how its performance is better than the last model. The RMSE and MAD continue to drop in value and now sit at 81.98 and 66 respectively. The adjusted r-squared increases slightly to capture 78.6% of the variation of the index that can be explained with the time and tax variables.

The addition of the lags helps diminish the intensity of the seasonal patterns from the tax data set but is still consistently over-forecasting for the second quarters. (Figure 4) The p-value is 0.41, which is above the alpha of 0.05, so it still shows no relationship between the time variable and



the index. It also turns to a negative value that decreases more with each period. However, the overall p-value is lower than the alpha, so the other variables still have a significance in the regression.

Lastly, the fourth model improves on all the past regressions by adding seasonal dummy variables. This considers the seasonal aspects by adjusting the forecasted index to a specific quarter of the year. The equation to predict the index carries all the same variables as before but adds three more for each quarter except for the fourth one. The last one is accounted for if all the other ones are counted as zero. When measuring how close the forecasted values came to the observed, the RMSE and the MAD both decreased to 79.96 and 63. The adjusted r-squared increased slightly to 78.94%.

The addition of the dummy variables minimized the intensity of the seasonal peaks from before and the data is closer to the observed. However, it does overreact to any shift in the observed index over the time-period. The overall p-value is lower than the alpha, showing there is significance in at least one of the

independent variables used. A closer look shows that the time and dummy variables are inconclusive in finding a relationship with the index. (Figure 5)

The next step in my forecasting process is to evaluate the performance of each model. To do this, I set up a table that displays all the models with their equations and compared each of the indicators of performance side by side. (Figure 6) I have already discussed the trends of how the RMSE and MAD have been getting smaller with each additional model.

Figure 5

OVERALL FIT					
Multiple R	0.897005045	AIC	874.77873		
R Square	0.80461805	AICc	876.82418		
Adjusted R Sc	0.789421676	SBC	895.45847		
Standard Errc	83.43793281				
Observations	98				

ANOVA					
	df	SS	MS	F	p-value
Regression	7	2580328	368618.28	52.948029	3.253E-29
Residual	90	626569.98	6961.8886		
Total	97	3206898			

	coeff	std err	t stat	p-value	lower	upper
Intercept	129.8150	32.780836	3.9600893	0.0001496	64.690184	194.93989
Group 1	-0.9895	0.7491849	-1.3207843	0.1899219	-2.4778981	0.4988749
Group 2	0.0039	0.001339	2.9099506	0.0045546	0.0012362	0.0065565
Group 3	0.0046	0.0013266	3.4398529	0.0008841	0.0019278	0.0071989
Group 4	0.0043	0.0013353	3.2530833	0.0016082	0.001691	0.0069965
Group 5	37.9880	29.435036	1.2905706	0.2001583	-20.489843	96.465829
Group 6	-5.4216	34.315192	-0.1579948	0.8748148	-73.594741	62.751496
Group 7	-10.9012	33.140163	-0.3289434	0.7429627	-76.739955	54.937481

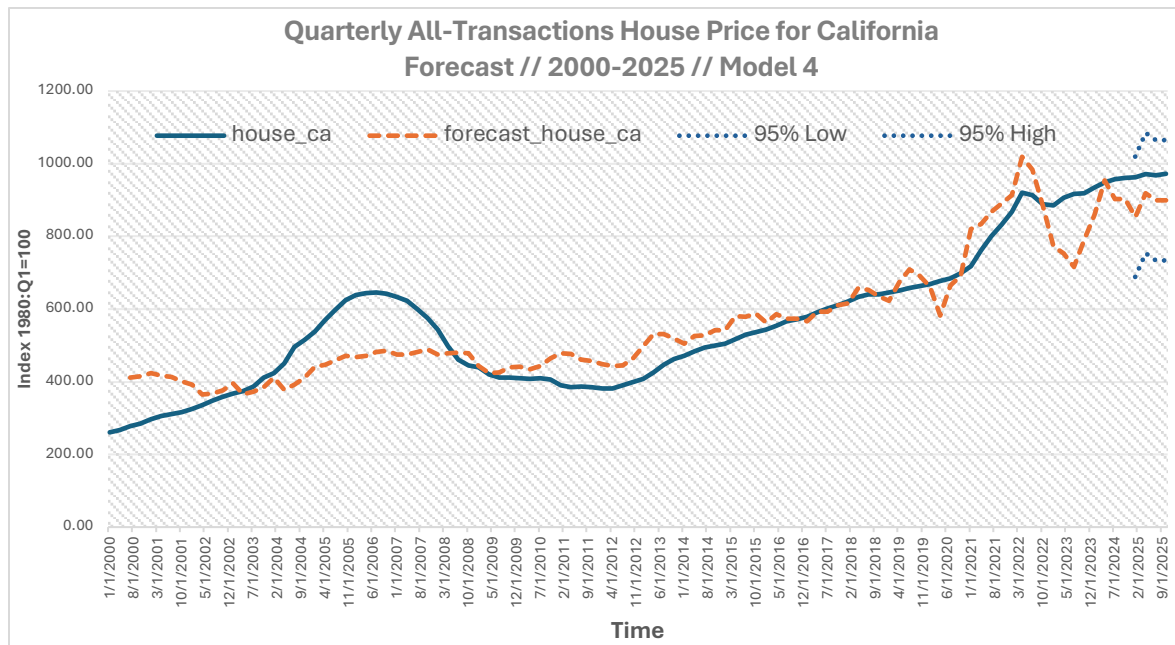
To calculate the Theil's U indicator, I made a Naïve model with the house index data and calculated the RMSE to compare with the model's RMSE. All the models did not perform as well when compared to the Naïve One as their values were above one. The fourth model performed the best out of the other regressions and will be used to forecast the next four quarters.

Figure 6

Forecast Equation	RMSE	AIC	MAD	Adjusted R ²	Theil's U
predicted house_ca = 292.8065 + 5.1380(t)	108.4637	941.2832	91.2591	0.6480	6.2948
predicted house_ca = 177.5214 + 0.8882(t) + 0.0043(tax_ca) + 0.0047(lag1_tax_ca)	89.7162	898.3370	73.3505	0.7502	5.2068
predicted house_ca = 145.7538 - 0.6021(t) + 0.0039(tax_ca) + 0.0040(lag1_tax_ca) + 0.0041(lag2_tax_ca)	81.9775	873.6631	66.0418	0.7858	4.7577
predicted house_ca = 129.8150 - 0.9895(t) + 0.0039(tax_ca) + 0.0046(lag1_tax_ca) + 0.0043(lag2_tax_ca) + 37.988(Q1) - 5.4216(Q2) - 10.9012(Q3)	79.9598	874.7787	63.0052	0.7894	4.6406

The layout for using the best model for the forecast involved adding a hold out year to test it on. That is a year with no observed data available to see how it performs on its own. For this to occur, I used the last four observed quarters to continue the tax variable and its lags. When calculating the prediction intervals, I multiplied the standard error and result from the t-distribution chart. Then I added and subtracted to get the range from each forecasted value. Next, I calculated the performance indicators to measure how it did compared to estimation sample. Lastly, I graphed the actual data with the forecasted values and prediction intervals to visually see how it performed. (Figure 7)

Figure 7



When measuring the performance of the fourth model, it under-forecasted from what was observed. The RMSE and MAD still outperform the other models in the measure of how close the forecast is to the actual value. The Theil's U is still above one which makes the Naïve model a better option than all the other models. The boundaries of the prediction intervals encompass the observed values to which concludes reasonable estimates for forecasting. In conclusion, the seasonal component of the tax variable made the forecast vary in its results. Each new variable introduced has slightly improved its performance but the Naïve One outperforms all of them.